MUHAMMAD SAMAD MAHAR

PHYSICS & COMPUTER SCIENCE UNDERGRADUATE

Victoria, BC | mmahar@uvic.ca | github.com/abdul-samad021 | LinkedIn

PROFILE

Physics and Computer Science undergraduate with experience in scientific Python, numerical analysis, data validation, course-based experimental physics, and technical team leadership. Building a reproducible detector-data pipeline, with additional team experience in physics-informed preprocessing and UI/UX for ML-assisted astrophysics; seeking research, scientific-computing, data, or software co-op opportunities.

EDUCATION

University of Victoria | BSc, Combined Major in Physics and Computer Science — **Expected May 2028**

Computer Science Co-op Program | Victoria, British Columbia | GPA: 7.77/9.00

Selected results: Intermediate Differential Equations 94%; Experimental Physics 92%; Electromagnetism 90%; Partial Differential Equations 88%; Probability & Statistics 87%; Numerical Analysis 85%

Current coursework: Classical Mechanics I; Quantum Mechanics I; Foundations of Computer Science (automata, computability, and complexity)

Additional coursework: Algorithms and Data Structures I-II; Operating Systems; Computer Architecture; Optics; Condensed Matter Physics; Computer-Assisted Physics

SELECTED PROJECTS

Uncertainty-Aware Cosmic-Ray Direction Reconstruction | Independent — **Aug 2026 - Present**

Python, NumPy, SciPy, pandas, Matplotlib, pytest, Ruff, Pierre Auger Open Data

- Built a checksum-recorded, reproducible data workflow and deterministic 1,000-event SD-1500 pilot cohort; implemented a physically constrained, uncertainty-weighted plane-front fit with SVD initialization, nonlinear optimization, and fit diagnostics.
- Validated known-truth and edge cases including angular quadrants, vertical and near-horizon directions, noisy timing, determinism, and degenerate geometry. Thirty-one repository tests pass; a typed event adapter and 64 contract tests pass locally, with repository integration in progress.

Exoplanet Detection Challenge Project | NASA Space Apps Challenge — **2025**

Physics-informed preprocessing, scientific advising, and UI/UX | Team project

- Contributed physics-informed data preprocessing before the team's machine-learning stage and advised on scientific interpretation and workflow choices for exoplanet transit detection.
- Developed UI/UX elements used to communicate the analysis workflow and candidate results during the hackathon; the team received a Rising Star Award.

COURSE LABORATORY EXPERIENCE

University of Victoria | Undergraduate Physics Course Laboratories

Supervised course experiments in measurement, optics, radiation detection, and electronics

- Completed supervised course experiments in optics and interferometry, diffraction, polarization, spectroscopy, radiation detection and counting, and analog electronics while following course safety procedures.
- Calibrated instruments; analyzed counting statistics and background effects; propagated uncertainty; applied curve fitting and residual analysis; compared results with theoretical models; and documented methods in technical laboratory reports.

LEADERSHIP EXPERIENCE

PragmaTech Solutions | Founder & Managing Director — **Jun 2024 - Present**

Hyderabad, Sindh, Pakistan

- Manage a 10-person technical team that has completed 300+ client software deliverables, coordinating requirements, task planning, technical contributions, quality review, and delivery in deadline-driven settings.
- Translate nontechnical requirements into scoped technical work and maintain clear client and team communication across concurrent deliverables.

TECHNICAL SKILLS

Programming and tools: Python; NumPy; SciPy; pandas; Matplotlib; Jupyter; Git; pytest; Ruff

Methods: numerical optimization; linear algebra; ODE/PDE modeling; statistical analysis; uncertainty propagation; data validation; scientific visualization; technical writing

RECOGNITION

NASA Space Apps Challenge: Rising Star Award - NASA International Space Apps Challenge, Victoria Local Event (team award, 2025) for the team's ML-assisted exoplanet-detection project (2025)

Additional awards: University of Victoria Entrance Scholarship ($5,000); GISUTECH Robotics Silver Medal for computer vision; INFOMATRIX World Finals Bronze Medal for iris recognition